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Simultaneous control of production, repair/replacement and preventive maintenance of deteriorating manufacturing systems

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ABSTRACT

This paper presents a method to find the optimal production, repair/replacement and preventive maintenance policies for a degraded manufacturing system. The system is subject to random machine failures and repairs. The status of the system is deemed to degrade with repair activities. When a failure occurs, the machine is either repaired or replaced, and a replacement action renews the machine, while a repair action brings it to a degraded operational state, with the next repair time increasing as the number of repairs increases as well. A preventive maintenance action is considered in order to improve the reliability of the machine, thereby reducing the amount of disruptions caused by machine failures. The decision variables are the production rate, the preventive maintenance rate and the repair/replacement switching policy upon machine failure. The objective of the study is to find the decision variables that minimize the overall cost, including repair, replacement, preventive maintenance, inventory holding and backlog costs over an infinite planning horizon. The proposed model is based on a semi-Markov decision process, and the stochastic dynamic programming method is used to obtain the optimality conditions. A numerical example is given to illustrate the proposed model, and a sensitivity analysis is considered in order to confirm the structure of the control policy and to illustrate the usefulness of the proposed approach.

1. Introduction

Recent years have seen considerable growth in interest in production planning and preventive maintenance control in manufacturing systems (see Dong-Ping (2009) and references therein). Such systems are subject to multiple uncertainties, including machine failures and repairs, explaining why repairable systems have attracted the attention of several researchers working in the field of deteriorating production systems (Soro et al., 2010). After a machine experiences a large number of failures, repair costs may become very high, and in such a situation, it becomes more reasonable to replace the machine with a new one. A preventive maintenance action could also be considered in order to increase the operating time of the machine and reduce the amount of production disruptions due to failures.

In a manufacturing environment, production activities, repairs, replacement and preventive maintenance actions are mutually interdependent. Gabriella et al. (2008) present a general overview of models and discuss some sectors of interaction between maintenance and production. Until now, no work has been conducted, which provides, in the same model, the production planning, the repair/replacement and the preventive maintenance policies for systems that deteriorate with age and the number of failures. In this paper, we combine the production, repair/replacement and preventive maintenance control activities of a stochastic manufacturing system subject to random breakdowns. Many articles have been written on the interactions between production, corrective maintenance, preventive maintenance, replacement and machine deteriorations without integrating them in a unified model as in this paper.

Preventive maintenance is usually planned and aimed at increasing the reliability of a deteriorating system or decreasing the operating costs of repairable systems (see Boukas and Haurie, 1990; Dellagi et al., 2007; Yuan et al., 2001; Kyriakidis and Dimitrakos, 2006; Karamatsoukis and Kyriakidis (2010)). Boukas and Haurie (1990) included an age-dependent machine failure rate and allowed the control to influence the jump rate from the machine operation mode to a preventive maintenance mode. They determined the production rate and maintenance rule, which minimize the total expected cost of a two-machine system.

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However, with the numerical scheme adopted in their work, it remains computationally difficult to obtain the optimal control of a large scale flexible manufacturing systems. To deal with this difficulty, Kenné and Boukas (2003) proposed a hierarchical control approach to determine the production and preventive maintenance policies for large scale manufacturing systems. A review of hierarchical approaches under uncertainties can be found in Sethi and Zhang (1994). Dehayem et al. (2009) used a hierarchical approach to develop a semi-Markov decision model in order to come up with a lower level determination of the optimal repair and replacement policy for a system that deteriorates with age, and is subject to damage failures. At higher levels, with that policy adopted, they derived a production plan for the system over an infinite horizon. However, there is no evidence that the structure of the control policy obtained at a lower level remains valid at a higher level of the hierarchy.

Yao et al. (2005) studied the joint preventive maintenance and production policies for an unreliable production system in which maintenance/repair times are non-negligible and random. Gharbi et al. (2007) assumed that failure frequencies can be reduced through preventive maintenance, and developed joint production and preventive maintenance policies depending on produced part inventory levels. In both authors' models, they assumed that maintenance actions always restore the machine to a perfect condition or does not change the failure rate of the system, and that the machine therefore performs similarly from one failure to the next. A more realistic assumption would be that the machine does not always return to a perfect condition following a corrective maintenance. Such a repair is called an imperfect repair (Liao, 2007).

While it is true that the imperfect repair model has been well documented for repair and replacement policies in manufacturing systems (Kijima et al., 1988; Love et al., 2000; Pérès and Noyes, 2003; Dimitrakos and Kyriakidis, 2008a, most available results unfortunately concern only cases where production and demand satisfaction are not taken into account. The authors considered that repair, replacement and preventive maintenance require a short time, and that they do not therefore significantly affect production activities (see Beichelt, 1992; Makis and Jardine, 1993).

If a machine is not initially new or is not new after each maintenance activity, it will clearly have different dynamics after each breakdown and repair. As stated by Pérès-Ocon and Torres-Castro (2002), under particular conditions involving exponential times, such a system can be modeled as continuous-time Markov chains or with Markov decision process (see Pavitsos and Kyriakidis, 2009). However, that is not the case when the probability distributions of operating and repair times follow general distributions, and where a semi-Markov process can then be used for modeling the system (see Dimitrakos and Kyriakidis, 2008b. Sloan (2008) developed a semi-Markov decision process model of a single-stage production system with multiple products and multiple maintenance actions. Such a model simultaneously determines maintenance and production schedules, while taking into account the fact that equipment conditions affect the yield of each product differently.

From previous and available studies, the research gap is that some of works on production planning took into account the fact that system deteriorates with age, but did not consider the cases in which parameters, such as repair time, change with the number of failures by increasing. One reason for this is that considering the variations in such parameters renders the problem more complex. Moreover, available studies did not simultaneously take into account the effect of a preventive maintenance control, repair/replacement and the effect of increasing repair times with the number of failures while controlling production. The mutual interdependency of those actions is therefore under-represented in the literature.

Our aim in this paper is to develop optimal strategies for manufacturing systems which take into account the deterioration of a machine in accordance with the number of failures (i.e., the repair time increases with the number of failures). Thus, repair activities depend on the machine's repair history and Markovian decision processes are no longer appropriate for the model of the control problem. The model proposed in this paper will provide the following three policies simultaneously:

- Repair/replacement switching policy based on the age of the machine and the number of failures above which the machine must be replaced if a failure occurs.
- Preventive maintenance policy based on the age of the machine at which preventive maintenance must be performed.
- Production policy based on the age of the machine and the stock level of finished goods.

The resulting simultaneous control approach consists in developing a semi-Markov decision process (SMDP) in order to determine optimal production, repair/replacement and preventive maintenance policies for the system. Such policies are determined in order to minimize inventory, backlog, repair, replacement and preventive maintenance costs over an infinite planning horizon.

The paper is organized as follows. In Section 2, we present the industrial context of the problem under study. In Section 3, we define the notations and assumptions used in the model. The problem statement is presented in Section 4. Numerical examples are presented in Section 5, and a sensitivity analysis is given in Section 6 to illustrate the usefulness of the proposed approach. Discussions are given in Section 7, and the paper is finally concluded in Section 8 with extensions.

2. Industrial context

The study presented in this paper has many applications, especially in the production industry. As stated by Badia and Berrade (to appear), many engineering systems are subject to the so-called unrevealed failures, which include failures that can only be detected through special tests, inspections or monitoring. Examples of such systems include seal machines, filling machines, machining centers, grinders, milling and many other machining tools. They generally have a large number of components (treadmill, ball screws, spindle heads, precision gear boxes, axis drive components, rotary tables, saddles, pallets and short, etc.), which stochastically deteriorate over time; thus causing the machines to also deteriorate over time. Very often, the parts of such machines are broken or damaged, even though the machine may be operational.

For example, worn nozzle, abrasives in pumped liquid, relief valve stuck, partially plugged or improperly adjusted, cavitations, worn bearing will not necessarily stop a liquid filling machine. A worn bearing, for example, will create a knocking noise, but still leave the machine operational. A machine failure may thus be attributable to one or more removable and repairable component. Over time, the number of components to be checked and repaired at failure increases, thus increasing the mean repair time as well.

Preventive interventions, which are intended to reduce breakdown risk and maintain the smooth operation of machines, provide the opportunity to clean and adjust valves, control stroke and flow injection, replace worn nozzles with properly sized ones, etc.

The model presented in this paper is suitable for repairable manufacturing systems characterized by deterioration due to production and failures. The overall production system is renewed through a replacement after a sequence of successive failures and repairs, in order to allow it to maintain a competitive advantage.

3. Notations and assumptions

Throughout this article, we use the following notations and assumptions:

3.1. Notations

The following notations will be used to describe the proposed control model:

$x(\cdot)$	stock level
$n(t)$	number of failures at time t
$a(t)$	age of the machine at time t
$\xi(\cdot)$	stochastic process
d	demand rate
$q_{\alpha\beta}$	transition rate from mode α to β
Q	transition rate matrix
$g(\cdot)$	instantaneous cost
c^+	holding cost per unit of item per unit of time
c^-	backlog cost per unit of item per unit of time
c_0	replacement cost
c_r	repair cost
$J(\cdot)$	total cost
$v(\cdot)$	value function
ρ	discount rate
$u(\cdot)$	production rate of the manufacturing system
$\omega(\cdot)$	preventive maintenance rate of the machine
u_{max}	maximal production rate of the manufacturing system
$\underline{\omega}$	minimal preventive maintenance rate
$\overline{\omega}$	maximal preventive maintenance rate
$s_n(\cdot)$	age of the machine for a systematic replacement at a given number of failures n
$N_{systrep}$	number of failures for a systematic replacement of the machine at failure

3.2. Assumptions

The mathematical model in this analysis is based on the following assumptions:

1) The planning horizon is infinite.
2) Failures of the machine are instantly detected and repaired (corrective maintenance).
3) The shortage cost depends on the shortage quantity and time (average value ($/unit of time)).
4) The holding cost depends on the mean inventory level (average value ($/unit of time)).
5) The cost of any replacement activity after failure is greater than the cost of a corrective or preventive maintenance activity. In addition, the mean replacement time, (i.e., the mean time it takes to replace the machine after a failure) is less than the time it takes to repair the machine when a failure occurs.
6) The mean time to repair increases with the number of failures and the costs also increase with the number of failures.
7) The mean time to replace the machine (i.e., the mean time the machine spends in operation before replacement at failure) is longer than the mean time to failure.

4. Problem formulation

In this section, we describe a stochastic model for a deteriorating failure-prone machine producing one type of product to meet customers' demand. The repairable machine goes through a finite number of successive failures before it is replaced with a new one. Therefore, we can assume that there is a maximum number of failures N that the machine can experience before replacement in order to guarantee the feasibility of the system. Thus, the number of failures $n(t) \in \{0,1,\ldots,N\}$. For a given number of failures, the inventory/backlog of the system, $x(t)$, evolves according to the following differential equation:

$$\dot{x}(t) = u(t) - d, \quad x(0) = x \tag{1}$$

where $x$is the initial stock level and $u(t)$ is the production rate of the machine at time t, with $x(t) > 0$ representing an inventory surplus and $x(t) < 0$ a backlog of $-x(t)$ parts. In the operational mode of the machine, we assume that $0 \le u(t) \le u_{\max}$ where u_{max} is its maximum production capacity. As the machine produces parts, its age increases. The machine age after the nth failure is given by

$$a_n(t) = a(t - t_n) + \theta_n, \quad t_n \le t < t_{n+1}, \quad t_0 = 0, \quad \theta_0 = 0 \tag{2}$$

where t_n is the instant of the nth failure and θ_n is the virtual age of the machine after the nth repair given by the following expression:

$$\theta_n = \psi \times a(t_n - t_{n-1}) + \theta_{n-1}, \quad n \ge 1 \tag{3}$$

where ψ is known as the repair intensity, and is assumed to be constant in this paper ($0 \le \psi < 1$). Note that $a_n(t)$ is used to specify that n failures have already occurred and express the historical relationship between the age and the production of the machine. A repaired machine is as good as new if $\psi = 0$; otherwise, its virtual age increase with the number of failures (i.e., $\theta_0 < \theta_1 < \theta_2 < \cdots < \theta_n < \cdots$). The function $a(t)$ is an increasing function of the number of produced parts since the last restart of the machine. It is described by the following differential equation:

$$\dot{a}(t) = ku(\cdot), \quad a(T) = 0 \tag{4}$$

where T is the last restart time of the machine (i.e., $T = 0, t_1, t_2, \cdots$), k is a given positive constant.

Note that if n failures have already occurred, we will refer to the age by using $a_n(t)$ which is relates to the historical evolution of the machine.

The machine modes can be classified as operational, denoted by 1; under repair, denoted by 2; under replacement, denoted by 3, and under preventive maintenance, denoted by 4. The mode of the machine at time t is then given by $\xi(t) \in \Omega = \{1,2,3,4\}$ such that

$$\xi(t) = \begin{cases} 1 & \text{if the machine is operational} \\ 2 & \text{if the machine is under repair} \\ 3 & \text{if the machine is under replacement} \\ 4 & \text{if the machine is under preventive maintenance} \end{cases}$$

The machine may randomly be at any of the four modes over an infinite horizon, as described in Appendix A, where the transition diagram and details on transition rates are provided.

Let $s_n(x) \ge 0$ be the age at which the machine is replaced if a failure occurs. The replacement switching age $s_n(x)$ is assumed to be a control variable for a given stock level x. Moreover, since each manufacturing system that deteriorates with age is to be replaced in the long term, we can assume that there exists an upper bound on the replacement age, M_{up}, which is very large (compared to $s_1(\cdot)$), and beyond which the machine is systematically replaced.

For any state of the system$(x(t),\xi(t),a(t))$, we can identify the action to be undertaken at the nth failure as follows:

– If $n < N$ and $a(t) < s_n(x)$, then the machine is to be repaired
– Else the machine is replaced due to one of the following three situations:
  a. The number of failures exceeds N

b. The age of the machine reaches $s_n(x)$
c. The age of the machine exceeds M_{up}

The corrective maintenance of the machine is not perfect in the sense that the repair time increases with the number of failures. After corrective maintenance, the age is reset to the virtual age θ_n. At the next failure, it will take much more time to repair the machine. Thus, repair activities depend on the machine's repair history, and the system is thus described by a semi-Markov decision process (SMDP). We shall therefore refer to $Q(\cdot)$ as the transition matrix of the semi-Markov chain$\xi(\cdot)$. The control variables are the age $s_n(\cdot)$ after which the machine should be systematically replaced at the next failure, the production rate $u(\cdot)$ and the preventive maintenance rate $\omega(\cdot)$. Hence, the set of the feasible control policies $\Gamma(\alpha)$, including $s_n(\cdot)$,$u(\cdot)$ and $\omega(\cdot)$, depends on the machine mode $\alpha \in \Omega$ and is given by

$$\Gamma(\alpha) = \{(s_n(\cdot),u(\cdot),\omega(\cdot)) \in \Re^3, s_n(\cdot) \geq 0, 0 \leq u(\cdot) \leq u_{max}\mathrm{Ind}\{\alpha = 1\}, \underline{\omega} \leq \omega(\cdot) \leq \overline{\omega}\} \tag{5}$$

The instantaneous cost associated with the considered planning problem is given by

$$g(\xi(t),a,x,n,s_n,u,\omega) = c^+x^+ + c^-x^- + c_r\mathrm{Ind}\{\xi(t)=2\} + c_0\mathrm{Ind}\{\xi(t)=3\} + c_m\mathrm{Ind}\{\xi(t)=4\} \tag{6}$$

where constants c^+ and c^- ($ per parts per unit of time) are used to penalize inventory and backlog, respectively; $x^+ = \max(0,x)$, $x^- = \max(0,-x)$, c_r is the repair cost, c_0 is a replacement cost, c_m is the preventive maintenance cost (the cots c_r, c_0 and c_m are in $ per event); and the indicator function is defined as follows:

$$\mathrm{Ind}(\Theta(\cdot)) = \begin{cases} 1 & \text{if } \Theta(\cdot) \text{ is true} \\ 0 & \text{otherwise} \end{cases}$$

The production planning problem considered in this paper is the determination of the optimal control policy $(s_n^*(\cdot),u^*(\cdot),\omega^*(\cdot))$ minimizing the expected discounted cost, given by

$$J(\alpha,a,x,n,s_n,u,\omega) = E\left[\int_0^\infty e^{-\rho t} g(\alpha,a,x,n,s_n,u,\omega)dt \,\middle|\, x(0)=x, \xi(0)=\alpha, a(0)=a\right] \tag{7}$$

where ρ is a discount rate. The value function of the problem for a given number of failures n is defined by

$$v(\alpha,a,x,n) = \inf_{(s_n,u,\omega) \in \Gamma(\alpha)} J(\alpha,a,x,n,s_n,u,\omega) \tag{8}$$

The properties of the value function $v(\cdot)$ given by Eq. (8) are presented in Appendix B. The optimal control policy $(s_n^*(\cdot),u^*(\cdot), \omega^*(\cdot))$ denotes a minimizer over $\Gamma(\alpha)$ of the right-hand side of Eq. (B.1). This policy corresponds to the value function described by Eq. (8). Then, when the value function is available, an optimal control policy can be obtained, as in Eq. (B.1). However, an analytical solution of equations (B.1) is almost impossible to obtain. The numerical solution of the HJB equations (B.1) is a challenge considered insurmountable in the past. Boukas and Haurie (1990) showed that implementing Kushner's method can solve such a problem in the context of production planning. The numerical methods used to solve the proposed optimality conditions are also presented in Appendix B.

The system of equations (B.4) can be interpreted as the infinite horizon dynamic programming equation of a discrete-time, discrete-state decision process, as in Boukas and Haurie (1990) and Kenné et al. (2003) for production, repair/replacement and preventive maintenance planning problems. The discrete optimality conditions obtained, described by four equations (i.e., one equation for each mode $\alpha \in \Omega = \{1,2,3,4\}$) can be solved using either policy improvement or successive approximation methods. In this paper, we use the policy improvement technique to obtain a numerical solution of the considered planning optimization problem. The algorithm of this technique can be found in Kushner and Dupuis (1992).

In the next section, we provide a numerical example to illustrate the structure of the control policies.

5. Numerical example

This section provides a numerical example to illustrate the results obtained. The purpose is to show the effectiveness of the closed-loop production, preventive maintenance and repair/replacement switching policies for a deteriorating system, with a feedback on the number of failures, the age of the machine and the stock level. The instantaneous cost is the one described by Eq. (6) with values of parameters given later in this section. In this example, the mean time to repair the machine increases with the number of failures due to the repair time given by Eq. (A.1), presented in Appendix A, with $T_{21}(1) = 0.001$ and $r_b = 0.5, d_b = -3$ (i.e., the mean time to repair the machine at the first failure is small (less than a unit of time), and this increases with the number of failures n). The transition rate is $q_{21}(n) = T_{21}^{-1}(n)$ with $q_{21}(1) = 1000$.

The failure rate and the corresponding operational time are obtained from a Weibull distribution with scale parameter $\lambda = 0.05$ and a shape parameter $\alpha = 3$. In addition, $u_{max} = 0.4$ (parts/unit of time), $d = 0.25$ (*parts/unit of time*), $q_{31} = 10$ (number of replacements/unit of time), $q_{41} = 0.25$ (number of preventive maintenance per unit of time) with the factor $\psi = 0.01$. The other parameters are presented in Table 1.

The computational domain D is defined by

$$D = \{(x,a,n) : -5 \leq a \leq 15, \quad 0 \leq a \leq 100, \quad 0 \leq n \leq 20\}$$

The policy improvement technique is used to solve the system of four equations obtained from (B.4) for each mode of the machine. The results obtained are presented in this section to clearly illustrate the production, preventive maintenance and repair/replacement switching policies.

5.1. Production policy

The optimal production policy $u^*(a,x,\cdot)$, illustrated by Fig. 1(a), indicates the production rate, for a given number of failures n, for each stock level $x(t)$, and each age $a(t)$ of the machine.

To illustrate the optimal production policy, we use its boundary given by Fig. 1(b), and it is the optimal stock level $z_n^*(\cdot)$ such that if the stock level in the system $x(.)$ is less than the optimal stock level, the production should be at a maximum rate to reach the optimal stock level. Once the stock level in the system is equal to the optimal stock level, the production should be at the demand rate. If the stock level in the system exceeds the optimal stock level, then we do not produce. The optimal production policy recommends a production in zone A_u to reach the stock level $z_n^*(\cdot)$ and no production in zone B_u.

The optimal stock level is given by Fig. 2(a) for $n = 1,2,6,9$ and by Fig. 2(b) for $n > 9$.

In Fig. 2(a), when the machine has not yet had its first failure, the number of parts to be held in inventory is equal to zero, as it awaits the first failure. The corresponding is equivalent to the line $x = 0$ and zone A_u is inexistent. When the machine has experienced its nth failure, the number of parts to hold in inventory

Table 1
Parameters of the numerical example.

c_0	c_r	c_m	h_x	h_a	c^+	c^-	ρ	$\underline{\omega}$	$\overline{\omega}$
20,000	100	10	0.5	1	2	150	0.05	0.0001	0.1

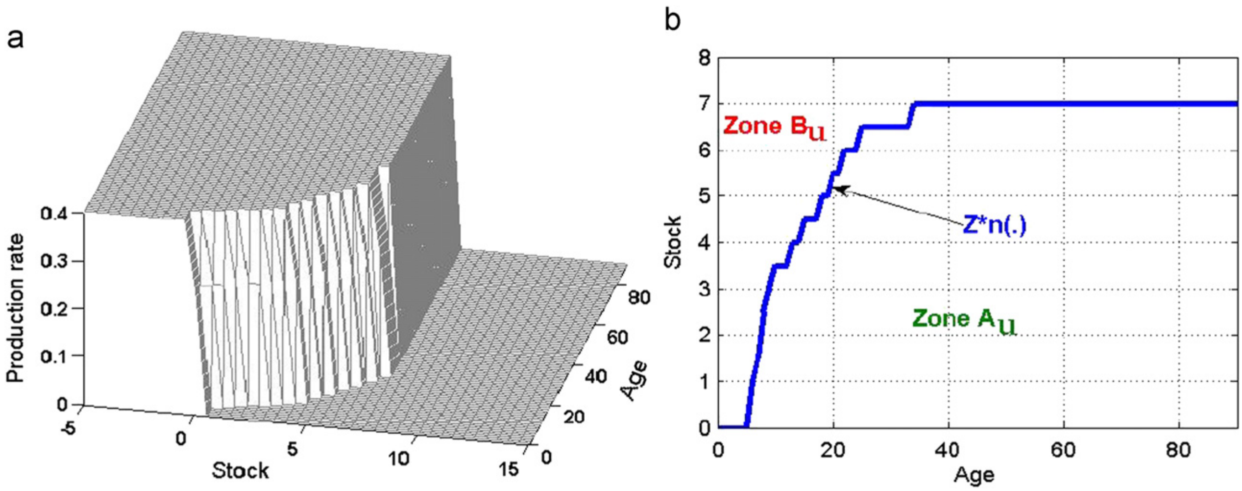

Fig. 1. Optimal production rate and its boundary: (a) $u^*(x,a,\cdot)$ and (b) $z_n^*(\cdot)$.

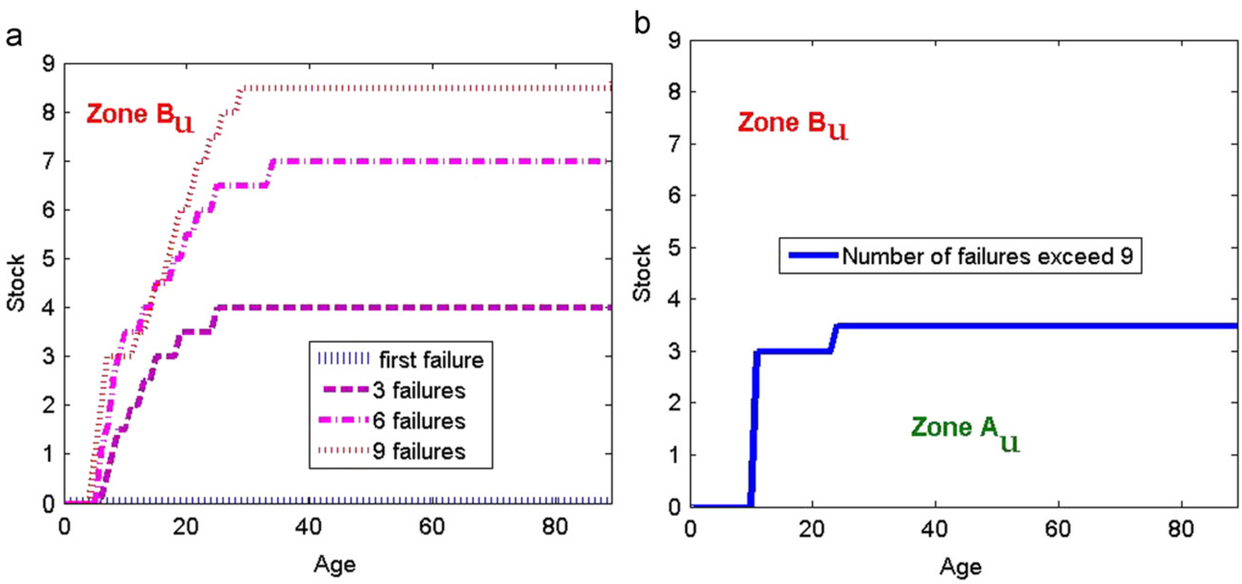

Fig. 2. Optimal stock levels $z_n^*(\cdot)$: (a) low number of failures; (b) high number of failures.

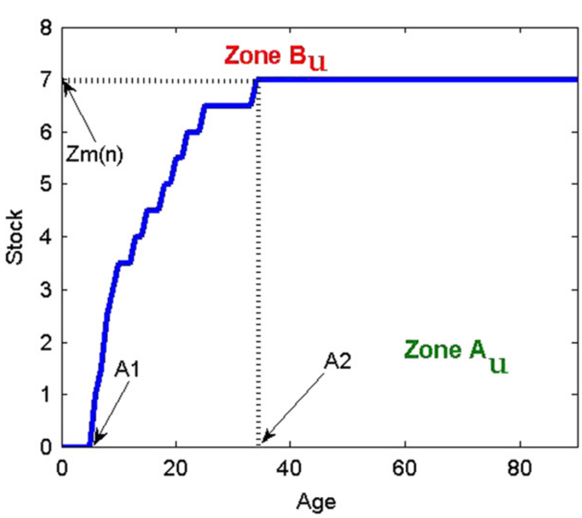

Fig. 3. Production policy $z_n^*(\cdot)$ parameters.

increases and attains a maximal value at $\max(z_n^*(\cdot))$; this value increases with the number of failures until a given maximal number of failures, as can be seen in Fig. 2(a) and (b). The corresponding maximal number of failures in this example is 9, as shown in Fig. 2(b). Moreover, when the number of failures exceeds 9, zones A_u and B_u are constant.

As illustrated by Fig. 3, the production policy is characterized by three parameters $A_1(n)$,$A_2(n)$,$z_m(n)$ such that

– If the age of the machine $a(t)$ is less than $A_1(n)$, the number of parts to hold in inventory is zero (i.e., $z_n^*(\cdot)=0$).
– If the machine's age $a(t)$ is such that $A_1(n) \le a(t) \le A_2(n)$, then the number of parts to hold in inventory must be brought from zero to $z_m(n)$.
– If the machine's age reaches $A_2(n)$, the stock must be maintained at level $z_m(n)$.

Fig. 4 gives the production policy for several failure numbers. The number of parts to hold in inventory increases as the age $a(t)$ and the number of failures n increase.

Fig. 4. Number of parts to hold in inventory for each number of failures.

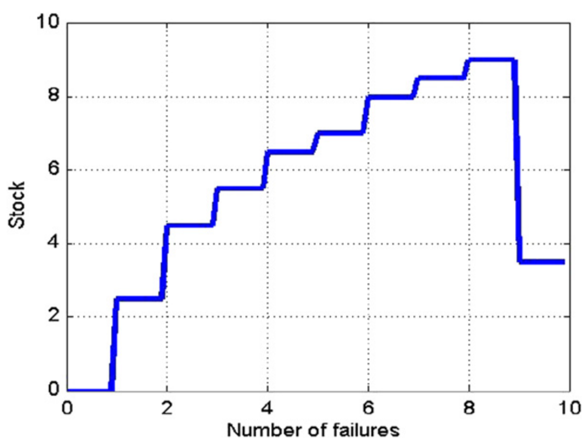

Fig. 5. Maximal number of parts $z_m(\cdot)$.

The maximal number of parts to hold in inventory for several numbers of failures presented in Fig. 5 increases from one failure to the next. This is logical because if the machine is to be repaired, it will take a greater amount of time. The optimal policy recommends holding a small number of parts in inventory after failure 9 because the machine will be renewed when failure 10 occurs.

Based on the results from Figs. 1–5, the production rate is given by the following machine failure-dependent hedging point:

$$u^*(1,a,x,n) = \begin{cases} u_{max} & \text{if } x(t) < z_n^*(\cdot) \\ d & \text{if } x(t) < z_n^*(\cdot) \\ 0 & \text{otherwise} \end{cases} \qquad (9)$$

with

$$z_n^*(\cdot) = \begin{cases} 0 & \text{if } a(t) \le A_1(n) \\ \in]0, z_m(n)[& \text{if } A_1(n) < a(t) < A_2(n) \\ z_m(n), & \text{if } a(t) \ge A_2(n) \end{cases}$$

It is important to note from Eq. (9) that when $a(t) > A_1(n)$, the stock level should be brought from 0 in order to reach $z_m(n)$ at $a(t) = A_2(n)$. The stock level is maintained at the value $z_m(n)$ when $a(t) \ge A_2(n)$.

5.2. Preventive maintenance policy

The optimal preventive maintenance policy obtained is presented in Fig. 6(a)–(c).

It shows that when the machine has not yet experienced its first failure (Fig. 6(a)), a preventive maintenance action is not recommended. Conversely, when the number of failures increases (Fig. 6(b) and (c)), machine age-dependent preventive maintenance actions are recommended. Throughout the rest of the paper, we use the boundary of the optimal preventive maintenance, namely Trace of PM, present in Fig. 7, to analyze the evolution of the optimal preventive maintenance policy. The boundary $T_\omega(x,n)$ of the optimal preventive maintenance policy divides the plan (x,a) into two zones:

- In zone A_ω, it is recommended to have the machine undergo preventive maintenance.
- In zone B_ω, the optimal preventive maintenance policy consists of not proceeding to a preventive maintenance of the machine.

The Trace of PM is presented in Fig. 8 for different numbers of failures.

When the machine has experienced less than 3 failures (indicated by arrow 1 in Fig. 8), zone A_ω is non-existent. There is no preventive maintenance action before the 3rd failure. Zone A_ω appears between failure 3 and failure 4, as indicated by arrow 2 in Fig. 8; it then increases, as indicated by arrow 3, and attains a constant value indicated by arrow 4, from failure 6 to failure 9.

The preventive maintenance actions are triggered according to a machine age-dependent policy described in Figs. 6–8. Such a policy requires that preventive maintenance be performed at a rate $\omega^*(\cdot)$ given by the following equation:

$$\omega^*(1,a,x,n) = \begin{cases} \underline{\omega} & \text{if } a(t) < T_\omega(\cdot); & \text{i.e., the zone } B_\omega \text{ is active} \\ \overline{\omega} & \text{otherwise;} & \text{i.e., the zone } A_\omega \text{ is active} \end{cases} \qquad (10)$$

where $T_\omega(\cdot)$ is the age limit for preventive maintenance before the nth failure of the machine.

5.3. Repair/replacement switching policy

The repair/replacement switching policy $s_n^*(x)$ obtained is presented in Fig. 9(a) and (b).

From Fig. 9(a), the age $s_n^*(x)$ over which the machine should be replaced if a failure occurs decreases with the number of failures and increases with the stock level. The age $s_n^*(x)$ is equal to 0 when the machine has experienced 9 failures. That means that while awaiting the 10th failure, the replacement age is 0, no matter the

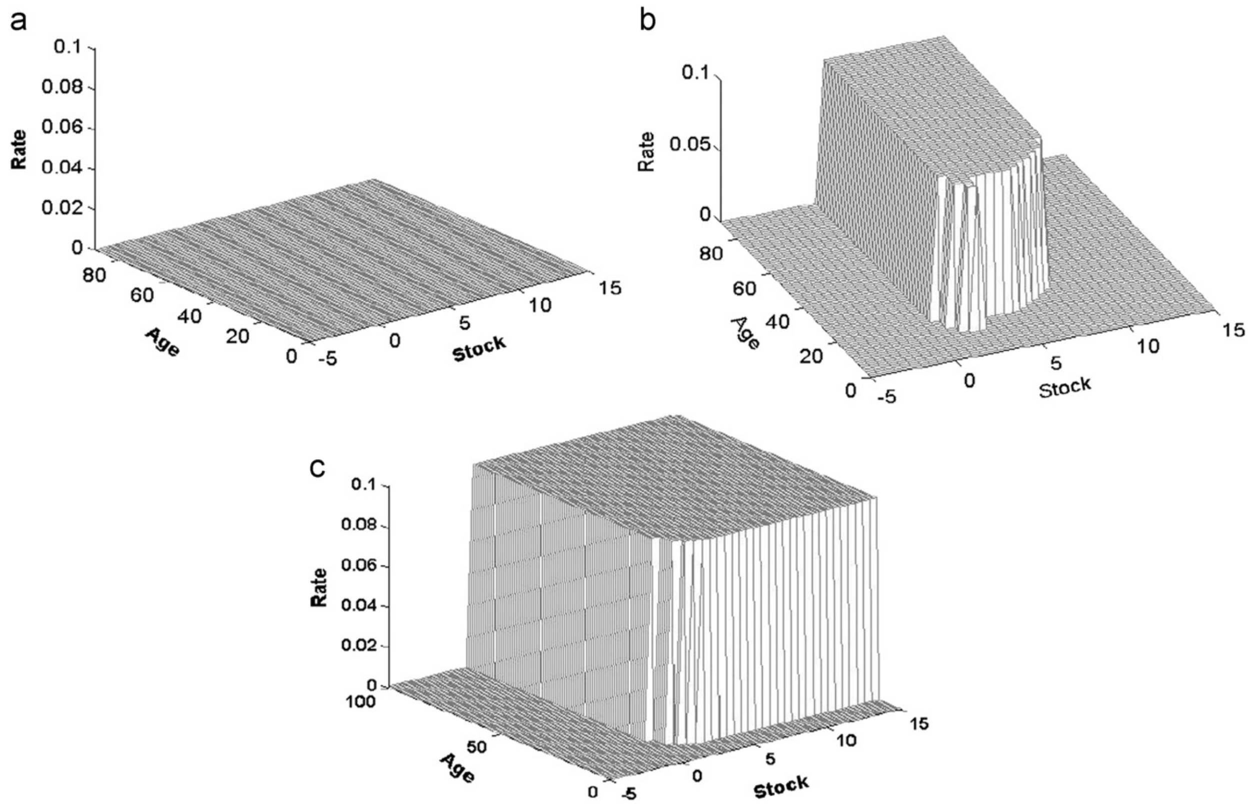

Fig. 6. Preventive maintenance policy $\omega^*(\cdot)$: (a) before 1 failure; (b) before 4 failures; (c) before 10 failures.

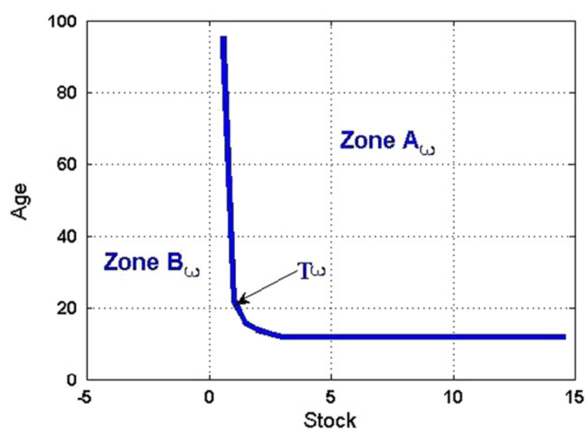

Fig. 7. Trace of the preventive maintenance policy $T_\omega(.)$.

stock level, and that the machine will experience a maximum number of 9 failures before systematic replacement at the next failure, no matter its age and the stock level. Hence, $N^*_{systrep}=9$.

The boundary of the repair/replacement switching policy, denoted by T_{sn}, is presented in Fig. 9(b). Such a boundary divides the plan (x,n) into two zones:

– In zone B_{Sn}, the machine is systematically replaced if a failure occurs.
– In zone A_{Sn}, the machine is repaired if a failure occurs above the replacement age $s_n^*(x)$.

Let $R_n(a,x)$ denote a function with value 1 if a repair action is undertaken after the $n$thfailure, and 0 if not. Based on the above results and with such a notation, the repair/replacement switching policy is given by the following expression:

$$R_n(a,x)=\begin{cases}1 & \text{if } a(t)<s_n^*(x);\ \text{i.e., the zone } A_{Sn} \text{ is active}\\ 0 & \text{otherwise; i.e., the zone } B_{Sn} \text{ is active}\end{cases} \tag{11}$$

with $s_n^*(x)$ given in Fig. 9(a).

In the next section, we will confirm the structure of the control policies obtained through a sensitivity analysis and illustrate the usefulness of the proposed approach.

6. Sensitivity and results analysis

A sensitivity analysis is presented in this section on cost parameters to confirm the structure of the obtained control policies given by Eqs. (9)–(11) and to illustrate the contribution of this paper. The sensitivity of the control policies is analyzed according to the variation of the cost of repair, the cost of replacement and the cost of preventive maintenance.

6.1. Variation of the repair cost

Fig. 10(a) illustrates the maximal number of parts to hold in inventory for four replacement cost values c_r=5, 10, 20 and 30. The results show that variations of the repair cost have no significant effect on the threshold value $z_m(\cdot)$, but that they affect the repair/replacement switching policy, as shown in Fig. 10(b). From Fig. 10(b), when the repair cost varies and takes the values 5, 10, 20 and 30, the machine is systematically replaced after the 13th, 10th, 7th and 6th failures, respectively. The tendency is then to replace the machine early as the repair cost increases. Thus, if the machine is to be replaced at the next failure, the number of parts to hold in inventory while awaiting that failure is not high.

The preventive maintenance policy, shown in Fig. 10(c), is significantly affected by variations of the repair cost. The area A_ω

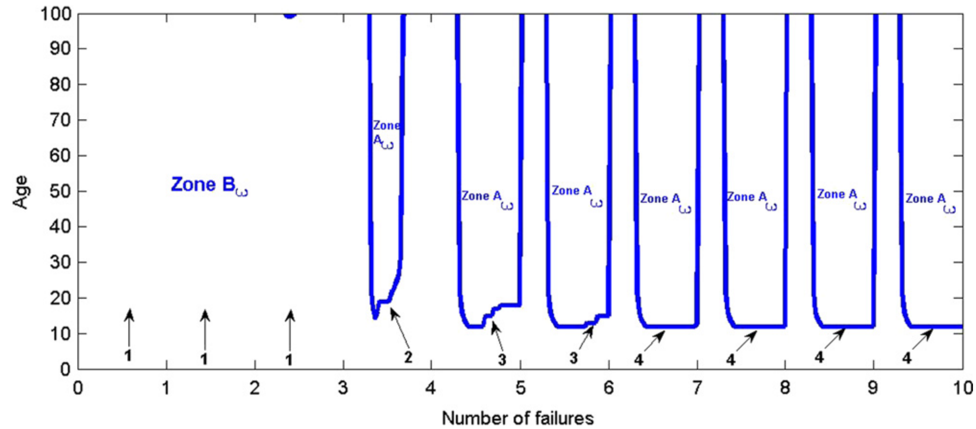

Fig. 8. Boundary of the preventive maintenance policy for several failures.

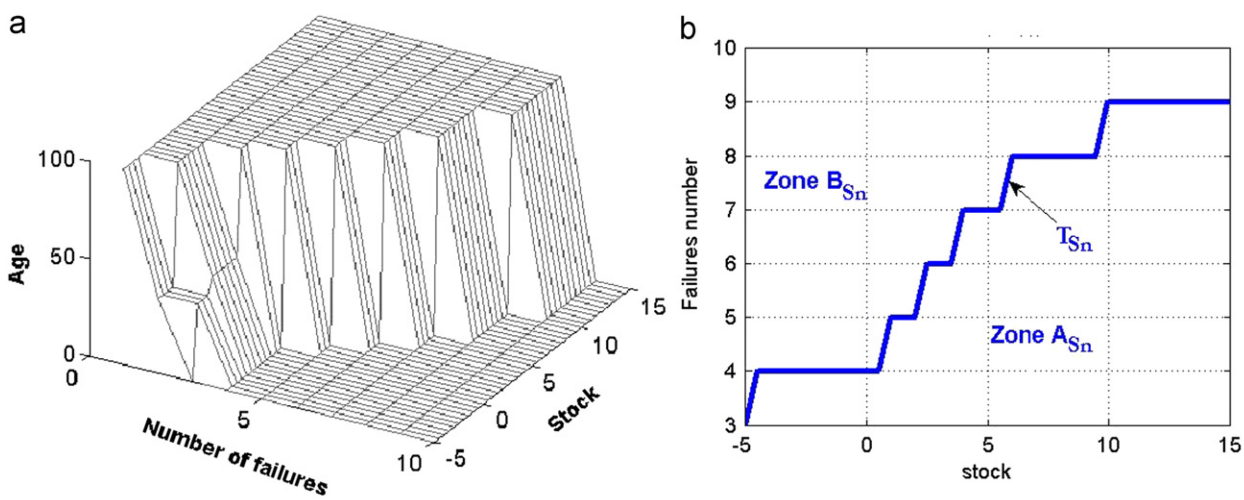

Fig. 9. Repair/replacement policy and its boundary: (a) $s_n^*(\cdot)$; (b) T_{sn}.

increases with the repair cost; that is, as repair costs increase, the preventive maintenance is recommended more often. It is clear from the above results that the repair costs significantly influence the optimal control policies obtained. The machine is replaced earlier (it experiences a lower number of failures) when the repair cost increases, and the optimal threshold depends on the number of failures the machine experiences and the preventive maintenance area increases.

6.2. Variation of the replacement cost

The variation of the replacement cost c_0 affects the optimal threshold stock level, as shown in Fig. 11(a). After a certain number of failures, the threshold level is significantly different from one replacement cost to the next. For a replacement cost value of 5000, from the 5th failure, there is no need to stock parts, while for a replacement cost value of 20,000, products should be stocked up to 9 failures.

Further, when the replacement cost is 5000, as illustrated in Fig. 11(b), the machine is systematically replaced at the 6th failure. From Fig. 11(c), we notice that there are only three, instead of four, curves given that the curve representing the value $c_0 = 5000$ is not illustrated. This is because no preventive maintenance is recommended when the replacement cost is 5000. This observation is logical in that since the replacement is not expensive, it is better to replace the machine with a new one as compared to sending it out for preventive maintenance and then expecting another failure.

6.3. Variation of the preventive maintenance cost

The variation of the preventive maintenance cost c_m has a relatively small effect on the production and repair/replacement policies, as shown in Fig. 12(a) and (b). However, it nevertheless significantly affects the preventive maintenance policy, as illustrated in Fig. 12(c), where we observe that for a high preventive maintenance cost value ($c_m = 300$), the recommended preventive maintenance area is reduced.

These conclusions on the influence of preventive maintenance cost on the production and preventive maintenance policies approximate those obtained by Gharbi et al. (2007), who observed, when formulating an analytical model for the joint determination of an optimal age-dependent buffer inventory and preventive maintenance policy in a production environment, that increasing preventive maintenance costs reduces preventive maintenance frequency and have no significant effect on the tendency of the production policy, as is seen in this paper.

7. Discussions

It is important to note that in many situations, the dynamics of several variables change after breakdowns as a result of the machine degradation phenomenon. As shown in the above results, taking into account some degradation with age after a machine failure leads to a policy comprising several threshold values, which increase from one breakdown to the next.

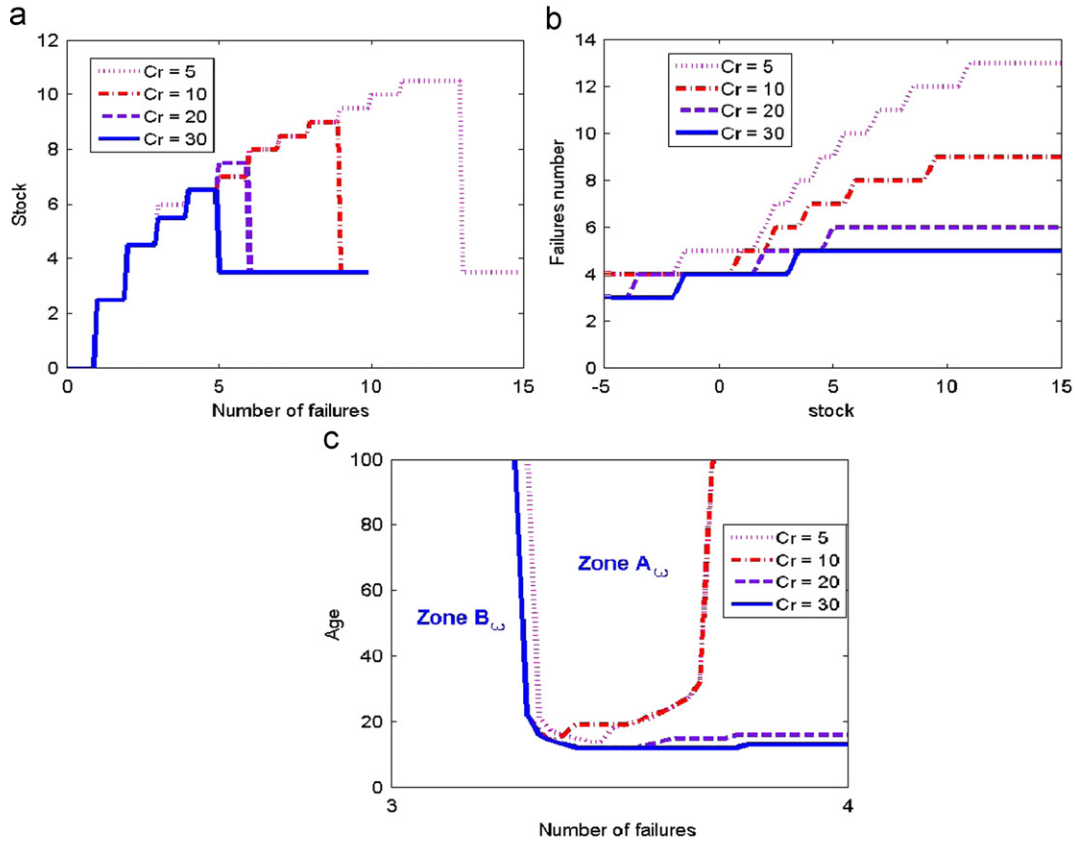

Fig. 10. Variation of the repair cost: (a) $z_m(n)$; (b) boundary of $s_n(\cdot)$; (c) $T_\omega(\cdot)$ for a given failure number n.

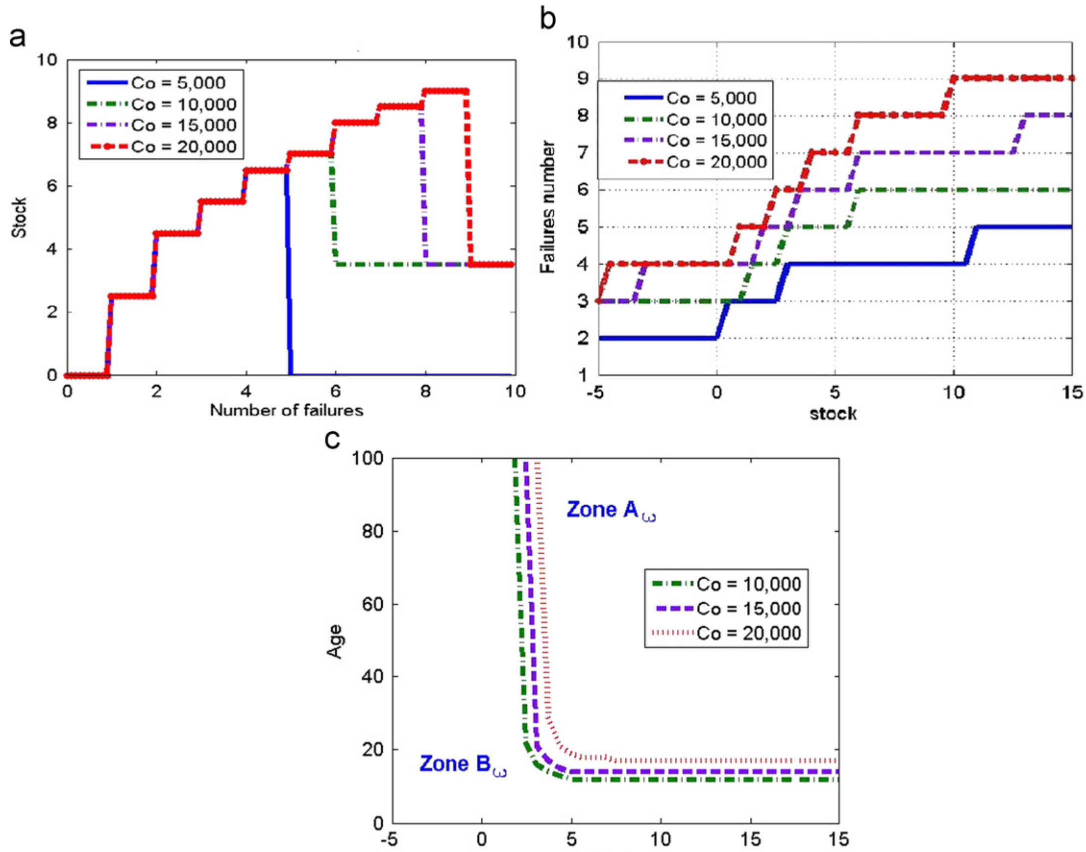

Fig. 11. Variation of the replacement cost: (a) $z_m(n)$; (b) boundary of $s_n^*(\cdot)$; (c) $T_\omega(\cdot)$ for a given failure number n.

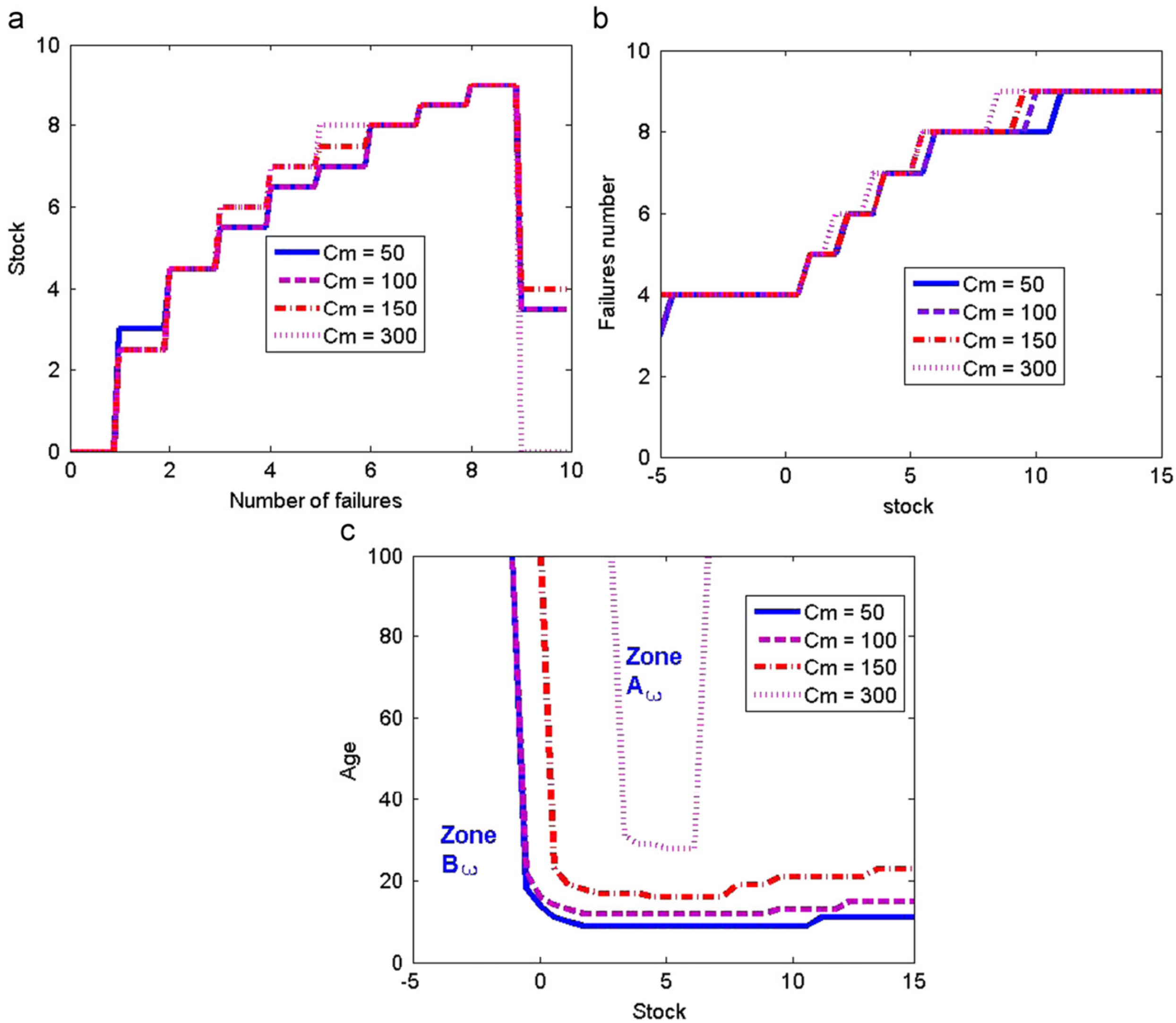

Fig. 12. Variation of the preventive maintenance cost: (a) $z_m(n)$; (b) $s_n^*(\cdot)$; (c) $T_\omega(\cdot)$ for a given failure number n.

The results obtained indicate that the optimal production, repair/replacement and preventive maintenance policies for the considered one-machine, one-part-type manufacturing system are characterized by a special structure. Such a structure is defined by

– three parameters for production (i.e., $z_m(n)$,$A_1(n)$,$A_2(n)$),
– two parameters for the repair/replacement switching policy (i.e. $s_n(x)$,$N_{systrep}$),
– one parameter for the preventive maintenance policy (i.e. $T_\omega(\cdot)$).

The overall control policy given by Eqs. (9)–(11) is completely defined by the values of six parameters $z_m(n)$,$A_1(n)$,$A_2(n)$), $s_n(x)$,$N_{systrep}$ and $T_\omega(\cdot)$ for production, repair/replacement switching and preventive maintenance policies.

For a given stock level, the structure of the optimal repair/replacement policy is an extension of that obtained by Makis and Jardine (1993) and by Love et al. (2000). As shown in section 5, their results appear to represent a particular case of the repair/replacement switching policy presented in this paper. Our results state that if at the nth failure, the age of the machine is above a certain critical value, the machine should be replaced; otherwise, an imperfect repair should be undertaken. The repair/replacement policy presented in this work is more realistic and appropriate for manufacturing systems because it takes into account the stock level in the system, the rate at which parts should be produced, and a preventive maintenance policy. Through the observations from the sensitivity analysis, it clearly appears that the proposed approach, based on a simultaneous control of production, repair/replacement and preventive maintenance strategies, provides more realistic results in the context of stochastic manufacturing systems.

8. Conclusion and extensions

This paper examined a production, repair/replacement and preventive maintenance planning problem in manufacturing systems. The objective of the study was to determine how to produce while the machine is in operation, when to replace versus repair the machine if a failure occurs, and when to perform a preventive maintenance, if any, in order to minimize the overall incurred cost. Our formulation included machine mode and control dependent jump rates, and allowed us to present the increasing probability of failures due to machine aging and increasing repair time resulting from imperfect repair activities. We used reset functions, which permit repair and preventive maintenance activities aimed at restoring the machine's age, and replacement, which provides a new machine. Thus, we extended the repair and replacement switching age control model to the case where production and preventive maintenance rates are controlled. Due to the fact that the repair activities of the machine depend on the repair history, we proposed the use of a semi-Markov decision process to determine optimal policies. We provided a numerical example, and through a couple of sensitivity analyses, we showed that the structure of the results obtained is maintained.

The study presented in this paper is limited to small system consisting of one machine producing one part type. For a more complex manufacturing system consisting of m machines, producing k different part types, the control policy depends on $6 \times m \times k$ parameters. The solution of HJB equations (as in Eq. (B.1)) is almost impossible for large m and k since the dimension of the numerical scheme to be implemented increases exponentially with the complexity of the system. For the aforementioned problem (i.e., for large m and k), a combination of the control theory (hierarchical approach) and the simulation-based experimental design, as in Gharbi and Kenné (2003), can be used to obtain a near optimal control policy, and this could result in the possible development of a more general model in production planning, which includes capacity limits, multi-product management and uncertainties on product demands.

Appendix A

The diagram in Fig. A1 illustrates the transitions between different modes of the machine.

The transition rates $q_{\alpha\beta},\alpha,\beta \in \Omega$, are described as follows:

- The failure rate $q_{12}(a_n(t))$ and the replacement rate $q_{13}(a_n(t))$ of the machine depend on its age $a(t)$, and are obtained from a machine age-dependent Weibull distribution.
- The machine is replaced at a constant rate q_{31}.
- The repair rate $q_{21}(n)$ is the inverse of $T_{21}(n)$. To define $T_{21}(n)$, let $t_{21}(n)$ be the repair time after the nth failure, and $T_{21}(n)$ the expectation of $t_{21}(n)$. The sequence $\{t_{21}(n), n=1,2,\ldots,N\}$ forms a non-decreasing arithmetic-geometric process with parameters $d_b \le 0$ and $0 < r_b \le 1$. We have for example $E(t_{21}(1)) = T_{21}(1) \ge 0$ and $T_{21}(1) = 0$ means that the repair time is negligible. As in Leung (2006), the non-decreasing repair time $T_{21}(n)$ is given by

$$T_{21}(n) = \frac{T_{21}(1)}{r_b^{n-1}} - (n-1)d_b, \quad n = 1,2,\ldots,N \tag{A.1}$$

- The transition rate from operational mode 1 to preventive maintenance mode 4 (i.e., $q_{14} = \omega(\cdot)$) is a control variable. The inverse of $\omega(\cdot)$ represents the expected delay between the decision to trigger preventive maintenance actions and the effective switch from operational to preventive maintenance mode (see Boukas and Haurie, 1990 for details). The following constraints hold for the preventive maintenance jump rate:

$$\underline{\omega} \le \omega(\cdot) \le \overline{\varpi} \tag{A.2}$$

where $\underline{\omega}$ and $\overline{\varpi}$ are the minimal and maximal preventive maintenance rates of the machine.
- The machine mode changes from preventive maintenance to operational mode at a constant rate q_{41}.

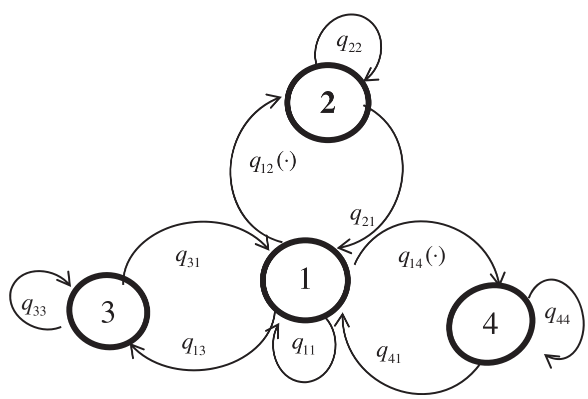

Fig. A1. State transition diagram.

Let $Q(\cdot) = (q_{\alpha\beta}(\cdot))$ denote a 4×4 matrix such that

$$q_{\alpha\beta}(\cdot) \ge 0, \alpha \ne \beta$$

$$q_{\alpha\alpha}(\cdot) = -\sum_{\alpha \ne \beta} q_{\alpha\beta}(\cdot)$$

The machine will be able to meet the demand rate d over an infinite planning horizon and reach a steady state, at a given machine age a_n, if

$$u_{max}\pi_1^n(a) > d \tag{A.3}$$

The machine age-dependent limiting probability of operational mode 1, for a given number of failures n, is shown to be $\pi_1^n(a) = 1/(1+(q_{21}(n)/q_{12}(a))+(q_{31}/q_{13}(a))+(q_{41}/\omega(\cdot)))$. Thus, $\pi_1^n(\cdot)$ depends on the decision variables as the transition rates are controlled through preventive maintenance. By controlling $\omega(\cdot)$, we act on the mean time before preventive maintenance. The system capacity is then described by a finite state semi-Markov chain that depends on the preventive maintenance policy.

Appendix B

The properties of the value function and how to obtain the Hamilton–Jacobi–Bellman (HJB) equations can be found in Kenné et al. (2003). Such equations describe the optimal control policies (optimality conditions) for production, repair/replacement and preventive maintenance problems. Regarding the optimality principle, we can write the HJB equations as follows:

$$\rho v(\alpha,a,x,n) = \min_{(u,\omega) \in \Gamma(\alpha)} \left(\min_{s_n \in \Gamma(\alpha)} \left\{ g(\cdot) + \frac{\partial v(\alpha,z,n)}{\partial a} \cdot \frac{da(t)}{dt} + \frac{\partial v(\alpha,z,n)}{\partial x}(u(\cdot)-d) + \sum_{\beta \ne \alpha} q_{\alpha\beta}(\cdot)[v(\beta,0,x,\psi(n)-v(\alpha,a,x,n)] \right\} \right) \tag{B.1}$$

where $\beta \in \Omega$,

$$\psi(n) = \begin{cases} 0 & \text{if } \xi(\tau^+) = 1 \quad \text{and} \quad \xi(\tau^-) = 3 \\ n+1 & \text{if } \xi(\tau^+) = 2,3 \quad \text{and} \quad \xi(\tau^-) = 1 \\ n & \text{otherwise} \end{cases}$$

$(\partial v(\cdot)/\partial x)$ and $(\partial v(\cdot)/\partial a)$ are the partial derivatives of the value function $v(\cdot)$ with respect to x and to a; and τ denotes the first jump time of $\xi(t)$.

Let us now develop the numerical methods for solving the optimality conditions given by Eq. (B.1) by the adaptation of Kushner's technique, as in Kushner and Dupuis (1992). Let h_x and h_a be the finite difference interval of variables x and a, respectively. The main idea of the approach consists in approximating $v(\alpha,a,x,n)$ by a function $v^h(\alpha,a,x,n)$ and the first-order partial derivatives of the value function $(\partial v(\cdot)/\partial x)$ and $(\partial v(\cdot)/\partial a)$, using the following expressions:

$$\frac{\partial}{\partial x}v(\alpha,a,x,n)=\begin{cases}\frac{1}{h_x}\left[v^h(\alpha,a,x+h_x,n)-v^h(\alpha,a,x,n)\right] & \text{if } u(t)-d\geq 0\\ \frac{1}{h_x}\left[v^h(\alpha,a,x,n)-v^h(\alpha,a,x-h_x,n)\right] & \text{otherwise}\end{cases} \tag{B.2}$$

$$\frac{\partial}{\partial a}v(\alpha,a,x,n)=\frac{1}{h_a}\left[v^h(\alpha,a+h_a,x,n)-v^h(\alpha,a,x,n)\right] \tag{B.3}$$

where $h=(h_x,h_a)$. With approximations given by Eqs. (B.2) and (B.3) and after some manipulations, the HJB equations can be rewritten as follows:

$$\begin{aligned}\rho v^h(\alpha,a,x,n)=\min_{(u,\omega)\,\in\,\Gamma(\alpha)}\Bigg(\min_{s_n\,\in\,\Gamma(\alpha)}\Bigg\{&\frac{g(\cdot)}{\Upsilon_h^{\alpha}(1+(\rho/\Upsilon_h^{\alpha}))}\\ &+\frac{1}{(1+(\rho/\Upsilon_h^{\alpha}))}\Bigg(P_x^+v^h(\alpha,a,x+h_x,n)+P_x^-v^h(\alpha,a,x-h_x,n)\\ &+P_a(\alpha)v_h(\alpha,a+h_a,x,n)\\ &+\sum_{\beta\neq\alpha}P^{\beta}(\alpha)v^h(\alpha,a,x,n)\Bigg)\Bigg\}\Bigg)\end{aligned} \tag{B.4}$$

where

$$\Upsilon_h^{\alpha}=|q_{\alpha\alpha}|+\frac{|u-d|}{h_x},\quad P_x^+(\alpha)=\begin{cases}\frac{u-d}{h_x\Upsilon_h^{\alpha}} & \text{if } u-d>0\\ 0 & \text{otherwise}\end{cases},$$

$$P_x^-(\alpha)=\begin{cases}\frac{d-u}{h_x\Upsilon_h^{\alpha}} & \text{if } u-d<0\\ 0 & \text{otherwise}\end{cases},\quad P_a(\alpha)=\frac{ku}{h_a\Upsilon_h^{\alpha}},\quad P^{\beta}(\alpha)=\frac{q_{\alpha\beta}}{\Upsilon_h^{\alpha}}$$

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